CS 577 Deep Learning
Homework 3: Attention and Transformer Forward

Your Name (Hawk username: `yourname`)

Fall 2026

Part 0: Setup and Axes

0.1 Environment and public tests

The notebook setup output was `Python 3.11.18`, `NumPy 2.4.6`, `PyTorch 2.15.0`, and `Matplotlib 3.11.2`; the device was `cpu` and the seed was `577`. I ran the public tests with `python -m unittest -v test_hw3_public.py`. The run executed 11 tests and failed with 11 errors because the starter functions raise `NotImplementedError`.

0.2 Axis and mask contract

TODO

Part 1: Hand Calculation

1.1 Unmasked attention

TODO

1.2 Causal attention

TODO

1.3 Scaling and stability

TODO

Part 2: Stable Masked Attention

2.1 Numerical stability and fully blocked rows

TODO

2.2 Hand-case agreement

TODO

2.3 Padding and causal masks

TODO

Part 3: Multi-Head Attention

3.1 Shape trace, attention maps, and outputs

Stage	Meaning	Shape
X	input features	TODO
XW_Q, XW_K, XW_V	projected features	TODO
Q, K, V	after splitting heads	TODO
S, A	scores and weights per head	TODO
Head outputs	weighted values per head	TODO
Merged outputs	concatenated head features	TODO
Y_{MHA}	after output projection	TODO

TODO

3.2 Independent PyTorch comparison

Case	Maximum output error	Maximum weight error
Unmasked self-attention	TODO	TODO
Causal self-attention	TODO	TODO
Rectangular cross-attention	TODO	TODO

TODO

3.3 Head count and head-output ablation

TODO

Part 4: Transformer Block

4.1 Shape trace and reference checks

TODO

4.2 Positions and reordering

TODO

4.3 Future perturbation and causal visibility

TODO

Part 5: Architecture Comparison

5.1 Encoder–decoder and decoder-only generation

TODO

Assistance and Generative-AI Disclosure

TODO